ModelIt!

Student Activity Pack

The Science of Sleep: Why Teens Can't Wake Up

Modeling Circadian Rhythms, Melatonin, and Sleep Architecture

Grade 12L1 | NGSS HS-LS1-3 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What time do you naturally fall asleep and wake up on weekends when you have no alarm? How different is this from school days?**

2. **Why do you think teenagers tend to stay up later than younger children or adults?**

3. **What do you think happens in your brain while you sleep — is it just resting or doing something active?**

4. **Do you think getting 6 hours of sleep from midnight to 6 AM is the same as 6 hours from 10 PM to 4 AM? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Light Exposure | Circadian Phase | Melatonin Level | Sleep Pressure | Cognitive Performance

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Natural Rhythm

Light: Natural daylight | No screens after 8 PM | Wake: Natural

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Blue Light Disruption

Light: 2 hours screen time before bed | Wake: 6:30 AM

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Sleep Debt Accumulation

Light: Screens until midnight | Wake: 6:30 AM | Repeat 5 days

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Caffeine Level — The concentration of caffeine in the bloodstream — blocks adenosine receptors to mask sleep pressure without actually reducing it, with a half-life of 5-6 hours

REM Sleep Proportion — The percentage of total sleep spent in REM stages — critical for memory consolidation and emotional processing, concentrated in the last 2-3 hours of sleep and disproportionately lost with early wake times

Cortisol Cycle — The daily rhythm of the stress hormone cortisol — normally peaks at wake time and drops at night, but becomes dysregulated with chronic sleep deprivation, contributing to anxiety and depression

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If sleep is so important for health and learning, why did evolution wire teenage brains to stay up late and wake up late — and why does society force you to fight your own biology every school morning? Use evidence from your simulation to support your answers.

1. **How does your model explain why you might feel exhausted on a school morning but wide awake at midnight?**

2. **Why is it not possible to simply 'train yourself' to be a morning person during adolescence?**

3. **What does the model predict about the effectiveness of catching up on sleep during weekends?**

4. **If a school district showed you data proving that later start times improve grades, attendance, and mental health, what argument could someone make against the change?**

5. **What are the limitations of modeling sleep with only five components?**

8 STEM Challenge: Design an Evidence-Based School Schedule

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YOUR ENGINEERING MISSION

Using your model data, design an optimal school schedule for your district that maximizes student cognitive performance while addressing practical constraints like transportation, athletics, and parent work schedules.

THE SCENARIO

Your school district is considering shifting high school start times from 7:30 AM to 8:45 AM based on sleep science. However, parents are concerned about after-school activities, bus scheduling, and supervision. The school board wants your team to present a data-driven proposal that addresses both the science and the logistics.

Engineering Design Thinking

1. WHAT SPECIFIC START AND END TIMES DOES YOUR SCHEDULE PROPOSE AND WHAT IS THE SCIENTIFIC BASIS?
2. HOW DOES YOUR DESIGN ADDRESS TRANSPORTATION, ATHLETICS, AND PARENT WORK SCHEDULE CONSTRAINTS?
3. WHAT MODEL EVIDENCE SUPPORTS YOUR CLAIM THAT THE NEW SCHEDULE WILL IMPROVE ACADEMIC OUTCOMES?
4. HOW WOULD YOU MEASURE THE EFFECTIVENESS OF YOUR PROPOSED SCHEDULE CHANGES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sleep scientists and chronobiologists research circadian rhythms and sleep disorders, working in hospitals, research universities, and pharmaceutical companies, earning $70,000-$140,000/year. Public health policy analysts who translate sleep research into school policy recommendations earn $55,000-$95,000/year.